An LLM-Based Exercise Prescription System for Digital Chronic Disease Management

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Abstract

Chronic diseases such as obesity and type 2 diabetes require sustained lifestyle modification and frequent behavioral feedback, yet providing safe, personalized and scalable exercise guidance remains challenging in real world settings, and prospective evaluations embedded in real world clinical workflows remain limited. We developed an expert-aligned large language model (LLM)–based Exercise Prescription System (EPS), supported by dedicated datasets that encode clinical and exercise-prescription knowledge for model alignment, and by domain-specific benchmarks that evaluate safety and medical competence. Across four benchmarks, EPS improved performance by 20–30% versus baseline models, with the best model approaching expert-level ratings. We then conducted an internal pilot study with 25 professional health managers and a single-blind randomized controlled trial in which 1,444 weight-loss participants (1,085 women; 359 men) and 40 glycemic-control participants (28 women; 12 men) were randomly assigned to human coaching, where exercise feedback was generated manually by health managers or EPS–human collaboration, where exercise feedback was generated by EPS and reviewed by health managers. Compared with the Human arm, the EPS–human arm achieved greater weight loss (1.40 (95% CI, 1.33–1.48) vs 1.20 kg (95% CI, 1.11–1.28); +16.67%; $\boldsymbol{P < 0.001}$) and a higher ≥2% weight-loss rate (62.94% (95% CI, 59.35–66.42) vs 54.27% (95% CI, 50.51–58.01); +8.67 percentage points; $\boldsymbol{P < 0.001}$), and yielded larger reductions in fasting glucose in glycemic control (0.98 (95% CI, 0.45–1.51) vs 0.30 mmol/L (95% CI, -0.05–0.65); +226.67%; $\boldsymbol{P < 0.05}$). Participants also reported high satisfaction, emotional stability and perceived encouragement from the EPS feedback.

These findings demonstrate that expert-aligned LLMs deliver clinically effective and scalable exercise guidance for real-world digital chronic disease management. Chinese Clinical Trial Registry identifier: ChiCTR2600118939

Keywords: Large language model, Exercise prescription, Chronic disease management, Randomized controlled trial

1 Introduction

Chronic diseases such as obesity and type 2 diabetes impose a major and growing global health burden [1, 2]. Recent epidemiological data indicate that over 800 million adults worldwide are currently living with obesity [3], while the number of individuals with diabetes is projected to surpass 850 million by 2050 [4], with diabetes expected to impose a cumulative global macroeconomic burden of INT$10.2 trillion during 2020–2050 [5]. These conditions drive substantial morbidity, premature mortality, and healthcare costs, and they are closely linked through shared, modifiable behavioral risk factors, including excess adiposity and physical inactivity [2]. Lifestyle-based interventions therefore remain foundational in prevention and treatment, with structured exercise and weight-management strategies improving cardiometabolic risk and, in selected individuals, enabling meaningful glycemic improvement and even remission [6–8]. Yet durable benefit requires sustained, long-term behavior change, and weight regain after initial loss remains common, reflecting the chronic, relapsing nature of obesity and the difficulty of maintaining new habits in real-world settings [9].

Despite strong evidence that intensive lifestyle programs improve cardiometabolic risk, sustained delivery in routine care remains difficult because effective support is labor-intensive and costly, and it scales poorly beyond specialized settings [10]. Digital health interventions aim to close this implementation gap by extending behavioural support outside the clinic and enabling more continuous, data-driven engagement [10]. In particular, Just-in-Time Adaptive Interventions (JITAIs) are designed to deliver the right type and amount of support at the right time by adapting to an individual's changing internal state and context using mobile and sensing technologies [11, 12]. However, current JITAI implementations often rely on hand-crafted decision rules and substantial expert input to specify tailoring variables, thresholds, and intervention options, which constrains adaptability across populations and limits scalable deployment [11, 13].

Recent advances in large language models (LLMs) create an opportunity to operationalize JITAI at scale by generating timely, personalized behavioral feedback from free-text user inputs. However, safety concerns remain central because LLMs may produce plausible but incorrect clinical statements, including under adversarial or ambiguous conditions, which limits their readiness for unsupervised health decision support [14–16]. Importantly, clinical evidence for LLM-enabled interventions remains sparse, with only a small number of prospective evaluations embedded in real-world workflows reported to date [17–19]. For example, a single-center randomized trial at hospital reception sites shows that nurse–LLM collaboration can improve patient

experience while retaining human oversight for uncertain outputs [17]. A multicenter pragmatic randomized trial further demonstrates that a co-designed patient-facing chatbot can support primary-to-specialist transitions by collecting preconsultation histories and generating structured referral reports [18]. To date, such prospective studies have largely focused on front-end access and intake steps such as reception, triage and referral preparation, whereas high-quality randomized evidence for sustained, personalized interventions in long-horizon chronic disease management remains limited. Even when domain-tuned medical LLMs achieve strong benchmark performance, expert assessments emphasize that multiple-choice scores do not fully capture factuality, calibration, and harm risk in real use, motivating more rigorous human and clinical evaluation [20–22]. Finally, many existing systems rely on broad public corpora that do not adequately cover specialized clinical subdomains and patient populations, and they rarely implement explicit JITAI decision rules, leaving a gap between general medical competence and context-aware, adaptive behavioral coaching [15, 23, 24].

Here we addressed this gap by proposing an LLM-based Exercise Prescription System (EPS) for digital chronic disease management that produces exercise feedback that is safe, professional, and personalized with high executability, while substantially reducing the workload of health managers through a human-in-the-loop workflow. Conceptually, EPS provides a practical paradigm for long-term, sustainable, and precise lifestyle intervention: the model generates individualized coaching from routine user inputs, and professionals focus on efficient review and risk control rather than end-to-end manual drafting. This paradigm is designed to generalize beyond obesity and diabetes to other chronic diseases in which sustained behavior change and frequent feedback are essential.

We developed and evaluated EPS with three primary contributions. First, we built dedicated resources for EPS, including a large-scale public corpus (D1) with over 80,000 exercise- and weight management–related medical question–answer pairs and a high-quality expert-curated dataset (D2) that captures common physical-activity concerns, expert preferences and professional judgment, together with four domain-specific medical benchmarks that quantify safety and medical competence in exercise and lifestyle counseling. Second, we introduced a two-stage alignment framework that integrated domain knowledge acquisition with expert preference alignment, combining supervised learning with Kahneman–Tversky Optimization and improving benchmark performance by 20–30%. Third, we demonstrated clinical and practical impact in real-world workflows through an internal pilot study with 25 professional health managers who evaluate the safety, guideline adherence and clinical usefulness of EPS-generated exercise feedback, and through a large-scale randomized controlled trial comparing human coaching (health-manager–written feedback) with EPS–human collaboration (EPS-drafted feedback reviewed by health managers). Across 1,444 weight-loss participants and 40 glycemic-control participants, EPS–human collaboration improved weight loss and fasting glucose reduction while maintaining high user-reported satisfaction and reducing health-manager workload, supporting a scalable paradigm for long-term personalized chronic disease intervention.

2 Results

2.1 Participant flow and baseline characteristics

The randomized controlled trials were conducted in WeChat-based online chronic disease management communities operated by Ping An Health and Technology Co., Ltd., with all participating health managers based in Shenzhen, China. Participants originated from the company's life-insurance user base. Most participants resided in mainland China and a small proportion resided in Hong Kong. Health managers assessed eligibility, obtained informed consent, and enrolled participants before 1:1 random assignment.

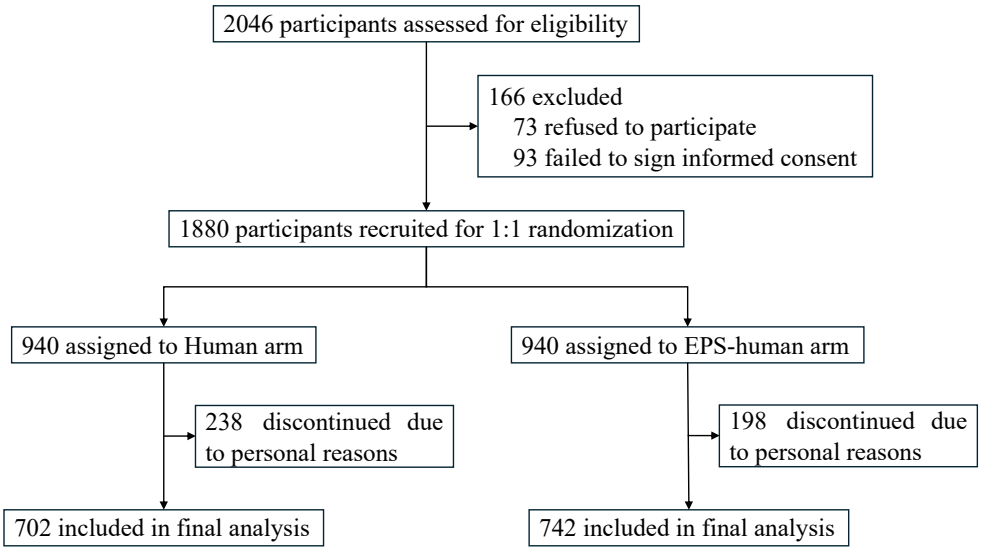

Fig. 1 CONSORT flow diagram for the randomized controlled trial in the weight-loss cohort. Participants were assessed for eligibility and, after informed consent was obtained, a subset was excluded. The remaining participants were randomized 1:1 to the Human arm or the EPS–human arm. Attrition occurred during follow-up as participants discontinued for personal reasons, and those who completed follow-up were included in the final analysis.

In the weight-loss cohort (Fig. 1), 2,046 participants were assessed for eligibility and 1,880 were recruited for randomization. Among those assessed, 166 participants were excluded, including 73 who refused participation and 93 who did not sign informed consent. Overall, 1,880 participants were randomly assigned (1:1) to the Human arm (n = 940) or the EPS–human arm (n = 940). In the final analysis, 702 participants remained in the Human arm and 742 remained in the EPS–human arm, with discontinuation due to personal reasons (238 and 198 participants, respectively). Baseline characteristics in the final analysis (Table 1), including age, sex distribution, BMI, and baseline weight, did not differ significantly between arms.

Table 1 Demographics of participating patients in the final analysis of the weight-loss clinical trial.

Characteristic	Total	Human	EPS–human	*P* value
No. of participants	n=1,444	n=702	n=742	
Age, years, mean ± s.d.	43.01±9.38	43.16±9.37	42.87±9.39	0.5644
Female sex, n (%)	1,085 (75.14%)	524 (74.64%)	561 (75.61%)	0.6723
BMI, mean ± s.d.	26.04±3.50	26.10±3.83	25.98±3.16	0.5269
Baseline weight (kg)	69.92±12.73	70.18±13.35	69.67±12.11	0.4430

Baseline characteristics are reported for participants included in the final analysis. Participants were randomized 1:1 to the Human arm or the EPS–human arm. Arms received identical program protocols except for the source of exercise feedback: feedback was generated manually by health managers in the Human arm and generated by EPS with health-manager review in the EPS–human arm. Continuous variables are presented as mean ± s.d., and categorical variables as n (%). P values are from two-sided Welch's t-tests for continuous variables and two-sided χ^2 tests for categorical variables. BMI, body mass index.

Table 2 Demographics of participating patients in the final analysis of the glycemic-control clinical trial

Characteristic	Total	Human	EPS–human	*P* value
No. of participants	n=40	n=21	n=19	
Age, years, mean ± s.d.	47.52±12.01	47.52±9.77	47.53±14.37	0.9995
Female sex, n (%)	28 (70.00%)	14 (66.67%)	14 (73.68%)	0.6286
BMI, mean ± s.d.	24.20±3.15	23.54±2.87	24.94±3.37	0.1669
Fasting glucose, mean ± s.d.	6.53±1.63	6.28±1.51	6.80±1.76	0.3313

Baseline characteristics are reported for participants included in the final analysis. Participants were randomized 1:1 to the Human arm or the EPS–human arm. Arms received identical program protocols except for the source of exercise feedback: feedback was generated manually by health managers in the Human arm and generated by EPS with health-manager review in the EPS–human arm. Continuous variables are presented as mean ± s.d., and categorical variables as n (%). P values are from two-sided Welch's t-tests for continuous variables and two-sided Fisher's exact tests for categorical variables. BMI, body mass index.

In the glycemic-control cohort (Fig. 2), 143 participants were assessed for eligibility and 78 were recruited for randomization. A total of 65 participants were excluded, including 23 who refused participation and 42 who did not sign informed consent. Ultimately, 39 participants were assigned to each arm. In the final analysis, 21 participants remained in the Human arm and 19 remain in the EPS–human arm, with discontinuation due to personal reasons (18 and 20 participants, respectively). Baseline characteristics in the final analysis (Table 2), including age, sex distribution, BMI, and baseline fasting glucose, did not differ significantly between arms.

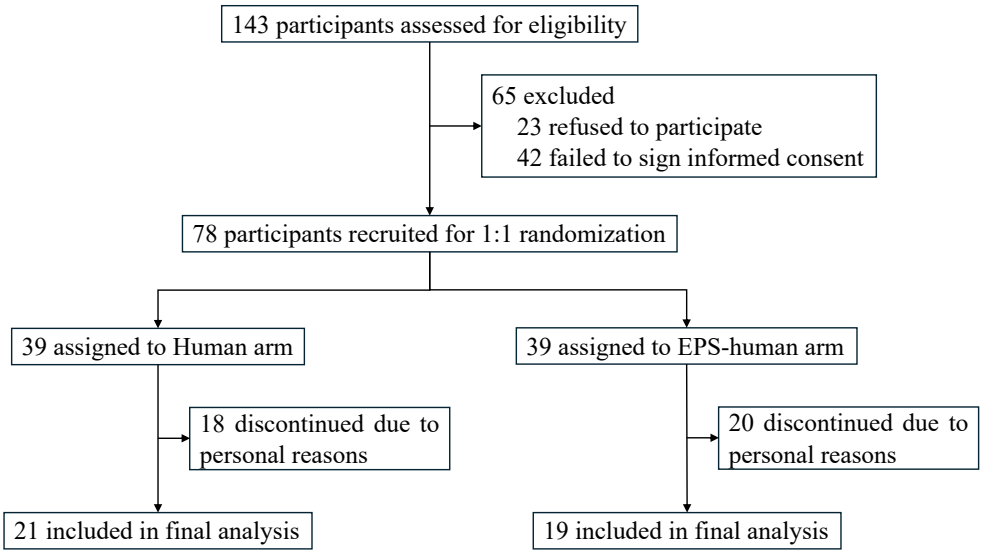

Fig. 2 CONSORT flow diagram for the randomized controlled trial in the glycemic-control cohort. Participants were assessed for eligibility and, after informed consent was obtained, a subset was excluded. The remaining participants were randomized 1:1 to the Human arm or the EPS–human arm. Attrition occurred during follow-up as participants discontinued for personal reasons, and those who completed follow-up were included in the final analysis.

2.2 Benchmark performance

2.3 Expert pilot evaluation

To quantify the incremental contributions of domain knowledge grounding (D1) and expert preference alignment (D2) in EPS, we conducted an ablation-style expert pilot evaluation (Fig. 3). Twenty-five professional health managers ($n = 25$) independently rated exercise-feedback outputs produced by three model variants: the Base model, EPS without D2 (D1-only), and EPS(D1+D2).

Across several key dimensions, EPS scores higher than EPS without D2, including correctness (2.48 (95% CI 2.16–2.80) vs 1.76 (95% CI 1.38–2.14), $P = 0.010$), completeness (2.52 (2.20–2.84) vs 1.68 (1.35–2.01), $P = 8.8 \times 10^{-4}$), unbiasedness (2.56 (2.27–2.85) vs 1.72 (1.30–2.14), $P = 0.0046$), and clarity (2.28 (2.06–2.50) vs 1.84 (1.58–2.10), $P = 0.012$).

Relative to the Base model, EPS without D2 shows higher ratings in consensus (2.00 (95% CI 1.66–2.34) vs 1.60 (95% CI 1.17–2.03), $P = 0.041$), correctness (1.76 (1.38–2.14) vs 1.24 (0.82–1.66), $P = 0.024$), clarity (1.84 (1.58–2.10) vs 1.28 (1.00–1.56), $P = 0.0058$), and empathy (1.64 (1.28–2.00) vs 1.08 (0.72–1.44), $P = 0.020$).

Together, these ablations suggest that D1 contributes to improved expert-rated quality relative to the base model, and adding D2 further strengthens EPS on multiple safety- and communication-relevant dimensions.

Table 3 Performance on four filtered medical benchmarks. Values are accuracy (%).

Group	Model	CMB (252)	CMExam (61)	MedMCQA (270)	MedQA (138)
Base models	DeepSeek-R1-8B	40.08%	39.02%	45.48%	31.96%
	Qwen3-8B	70.52%	60.66%	81.11%	66.09%
	DeepSeek-R1-14B	68.02%	62.13%	59.52%	36.45%
	Qwen3-14B	77.50%	76.39%	81.26%	68.12%
EPS	DeepSeek-R1-8B	54.14%	58.69%	63.74%	56.52%
	Qwen3-8B	84.76%	90.49%	88.41%	85.94%
	DeepSeek-R1-14B	84.40%	87.21%	85.04%	74.49%
	Qwen3-14B	86.75%	90.66%	89.74%	86.30%
Mainstream LLMs	ChatGPT-5	90.79%	93.77%	96.00%	97.68%
	DeepSeek-R1	89.13%	92.46%	95.63%	95.22%
	Gemini 2.5 Flash	86.75%	93.11%	94.67%	90.72%
	Grok 4 Fast	78.73%	88.52%	85.11%	85.51%

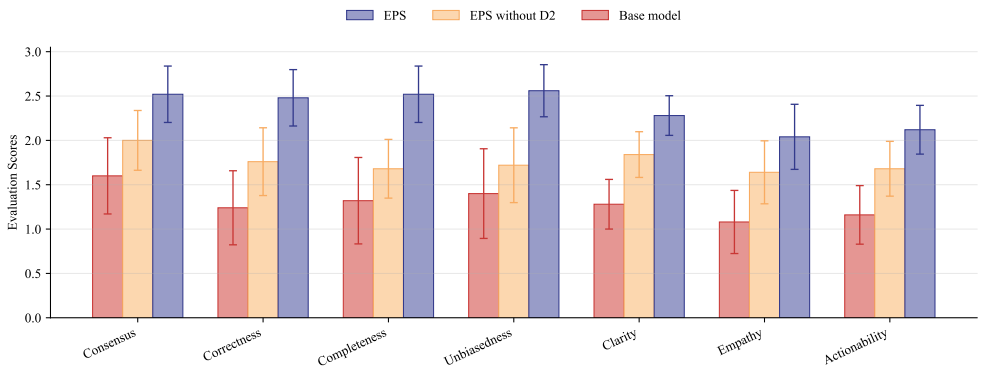

Fig. 3 Expert pilot evaluation. Grouped bar chart showing mean scores (0–3 scale) across seven dimensions (consensus, correctness, completeness, unbiasedness, clarity, empathy and actionability) for the Base model, EPS without D2, and EPS; error bars denote 95% confidence intervals.

2.4 Clinical effectiveness in a randomized controlled trial

We evaluated clinical effectiveness of EPS–human collaboration in two prospective randomized controlled trials embedded in WeChat-based chronic disease management communities, in which participants received the same 21-day intervention protocols and were randomized 1:1 to the Human arm or the EPS–human arm (Figs. 1 and 2). In the Human arm, exercise feedback was generated manually by health managers, whereas in the EPS–human arm, exercise feedback was generated by EPS and reviewed

by health managers before being delivered to participants. Baseline characteristics were balanced between arms in both cohorts (Tables 1 and 2).

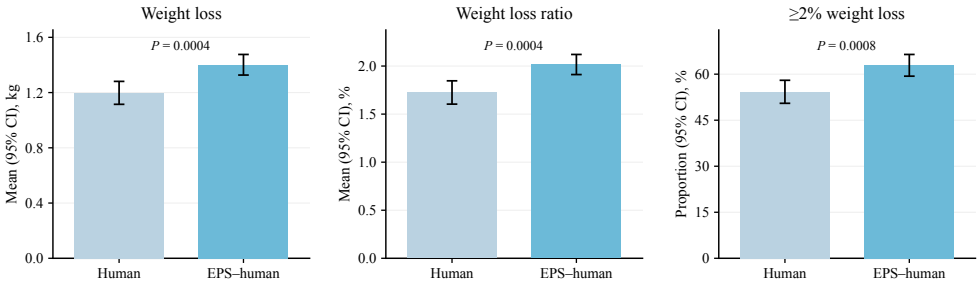

Fig. 4 Weight-loss outcomes in the weight-loss trial. Bars show arm means (weight loss in kg and percent weight loss) or the proportion achieving ≥2% weight loss. Error bars indicate 95% confidence intervals. P values are from Welch's t-test for continuous outcomes and a two-proportion z-test for the binary outcome.

In the weight-loss trial, the EPS–human arm achieved greater mean weight loss, the prespecified primary outcome, than the Human arm (1.40 ± 1.03 versus 1.20 ± 1.12 kg, $P = 0.0004$) (Fig. 4). The EPS–human arm also showed higher percentage weight loss, a prespecified secondary outcome ($P = 0.0004$) and a higher proportion of participants achieving ≥2% weight loss (62.94% versus 54.27%; $P = 0.0008$). Across the 21-day intervention, the EPS–human arm received more exercise-feedback instances than the Human arm (436 versus 263), consistent with higher feedback delivery intensity under EPS-assisted generation with human review. Among the 436 EPS-generated drafts, 11 (2.52%) required human edits before delivery, including 6 cases of language polishing, 2 cases of substantive content changes, 2 cases of structural adjustments, and 1 case of safety-related modification.

In the glycemic-control trial, participants in the EPS–human arm exhibited larger reductions in fasting glucose, the prespecified primary glycemic outcome, compared with the Human arm (0.98 ± 1.10 versus 0.30 ± 0.77 mmol/L, $P < 0.05$) (Fig. 5). The fasting glucose reduction ratio, a prespecified secondary glycemic outcome, was also greater in the EPS–human arm (12.60% (95% CI, 6.39–18.81) vs 3.38% (95% CI, −2.38–9.14), $P < 0.05$). A bootstrap sensitivity analysis yielded a consistent between-arm effect estimate for fasting glucose reduction (mean difference, 0.68 mmol/L, 95% bootstrap CI 0.17–1.34, $P < 0.05$) and for fasting glucose reduction ratio (mean difference, 9.22 percentage points, 95% bootstrap CI 2.28–17.74, $P < 0.05$). Across the 21-day intervention, the EPS–human arm received a greater number of exercise-feedback instances than the Human arm (87 vs 58), indicating higher feedback delivery intensity with EPS-assisted generation and human review.

Together, these results indicate that EPS–human collaboration improves short-term clinical outcomes in real-world digital chronic disease management under human oversight.

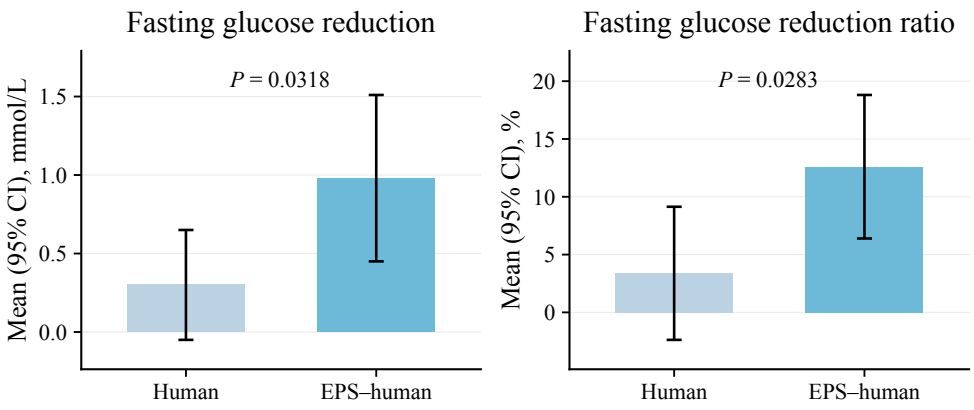

Fig. 5 Fasting glucose outcomes in the glycemic-control trial. Bars show mean fasting glucose reduction and mean fasting glucose reduction ratio in the Human and EPS–human arms. Error bars indicate 95% confidence intervals.

2.5 Participant-reported outcomes

Immediately after completion of the 21-day program, participants were invited to complete an end-of-program questionnaire about their experience of the exercise reviews. We excluded responses from participants who reported never having seen the exercise feedback (screening item 1), responses with implausibly short or long completion times, and "straight-lined" responses in which the same option was selected for all items. After quality control, we retained 344 valid questionnaires in the Human arm and 336 in the EPS–human arm.

Items 2–15 comprise 14 statements covering overall satisfaction and interaction experience, affective response after receiving feedback, perceived emotional stability and tone consistency, and perceived agreement/affirmation style of the feedback (Fig. 6). Each item was rated on a 7-point Likert scale (1 = strongly disagree to 7 = strongly agree). We summarize scores as mean (95% CI) and report between-arm comparisons using two-sided Welch's t-tests.

Across representative items, the EPS–human arm showed higher mean ratings than the Human arm, including overall satisfaction (item 2: 5.86 (95% CI 5.73–5.99) vs 4.51 (4.41–4.62), $P < 0.001$), pleasantness of interaction (item 3: 5.86 (5.73–5.98) vs 4.37 (4.26–4.49), $P < 0.001$), feeling happy after feedback (item 4: 5.86 (5.75–5.98) vs 4.26 (4.13–4.39), $P < 0.001$), perceived calmness and emotional stability (item 5: 5.94 (5.83–6.06) vs 4.38 (4.27–4.49), $P < 0.001$), steady tone under pressure (item 6: 5.78 (5.65–5.92) vs 4.22 (4.09–4.34), $P < 0.001$), tone consistency across scenarios (item 10: 5.63 (5.49–5.77) vs 4.16 (4.02–4.29), $P < 0.001$), and perceived praise/positive evaluation (item 14: 5.91 (5.79–6.04) vs 4.11 (3.97–4.25), $P < 0.001$).

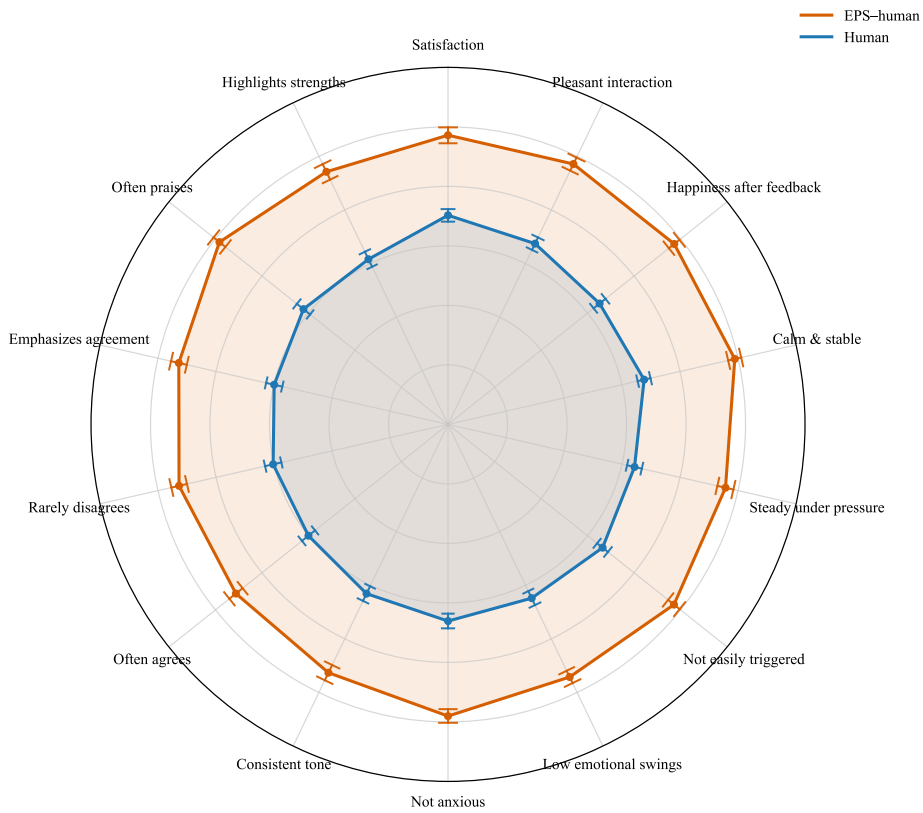

Fig. 6 Participant-reported experience in Phase 2. Radar plot summarizing mean questionnaire scores across 14 dimensions (items 2–15) for the Human and EPS–human arms. Each spoke corresponds to one questionnaire item, scored on a 1–7 Likert scale. Points indicate group means and radial error bars denote two-sided 95% confidence intervals estimated from completed responses.

3 Discussion

In this work, we developed an expert-aligned LLM–based EPS and evaluated it along a translational path from benchmark testing, to expert assessment, to prospective real-world deployment under human oversight. Across two pragmatic randomized trials embedded in routine WeChat-based chronic disease management communities, EPS–human collaboration produced modest but statistically significant improvements in short-term weight loss and fasting glucose reduction, while also increasing the intensity of feedback delivery. Together with favorable expert ratings and participant-reported experience, these results suggest that a carefully aligned LLM can serve as a scalable first-pass drafting system for personalized exercise coaching, when coupled with accountable human review.

Our findings have several implications for clinical AI and digital lifestyle medicine. First, they indicate that domain-specific alignment can meaningfully improve both

the medical quality and the behavioral quality of exercise guidance. Exercise coaching is not solely a factual task; it requires simultaneously respecting safety constraints and delivering behavior-change–relevant communication. We observed consistent gains in benchmark performance and expert ratings. The ablation results further indicate that successive alignment stages improve multiple quality dimensions, including correctness, completeness, empathy and actionability. Together, these findings suggest that aligning an LLM to structured exercise-prescription knowledge and expert preferences is a practical route to improving real-world coaching outputs beyond what a general-purpose base model can provide.

Second, the trials highlight a key operational mechanism by which LLM assistance may translate into clinical benefit: increasing the frequency and consistency of feedback delivery without removing human accountability. In both cohorts, participants in the EPS–human arm received substantially more feedback instances over the same intervention window, consistent with reduced marginal cost of drafting and faster turnaround under the collaboration workflow. This design reframes LLMs as tools that expand clinical capacity rather than replace clinicians [19]. This perspective is particularly relevant for chronic disease programs, where sustained engagement and repeated micro-interventions are essential, but staffing constraints often limit personalization and follow-up frequency.

Third, although the absolute effect sizes over 21 days were modest in the weight-loss cohort, the directionality across complementary endpoints (mean weight loss, percent weight loss, and the proportion achieving clinically meaningful loss thresholds) is consistent with improved adherence to prescribed activity and improved self-efficacy. In the glycemic-control cohort, larger mean reductions in fasting glucose and fasting glucose reduction ratio further suggest that EPS–human coaching can benefit metabolic outcomes, though this cohort was substantially smaller and thus warrants cautious interpretation and replication. The participant-reported outcomes provide convergent evidence that the mechanism is not purely informational: users rated EPS-generated feedback highly on satisfaction and perceived emotional stability, and they endorsed motivational elements such as encouragement and strengths-based framing [25]. These affective and relational properties are often difficult to scale in text-based coaching, yet they are central to sustained behavior change. Their presence in EPS outputs may help explain why an intervention that primarily changes the quality and cadence of text feedback can still produce measurable short-term clinical gains.

Fourth, our results contribute prospective evidence for LLM-enabled interventions in real-world digital health workflows, while also clarifying what is likely required for safe deployment in lifestyle medicine. Exercise prescription is a safety-critical domain: inappropriate intensity progression or advice that ignores symptoms and comorbidities can cause harm [26]. The expert pilot study suggests that, after alignment, EPS outputs are generally clinically plausible and safe in a controlled evaluation setting, but the trials were explicitly designed around human oversight, with health managers reviewing drafts before delivery. In practice, scalable deployment will likely require a spectrum of safeguards. This spectrum can range from human-in-the-loop review for high-risk cases to automated triage and structured constraint checks for routine cases, rather than a single binary choice between manual coaching and full automation.

Several limitations should be considered. First, intervention duration was short, and the trials primarily capture near-term behavior change and early physiological responses; longer follow-up is needed to assess durability, weight regain, and downstream outcomes. Second, the EPS–human arm delivered more feedback messages than the Human arm, which is an intended operational advantage but also a mediator; future work should disentangle whether improvements are driven mainly by increased cadence, improved content quality, or their interaction. Third, the glycemic-control cohort was small, and fasting glucose is sensitive to short-term behavioral fluctuations; larger samples, additional metabolic endpoints, and longer observation windows are required. Fourth, the setting was a specific platform and operational workflow; generalizability to other populations, languages, health systems, and coaching models remains to be established. Finally, while benchmarks and expert ratings provide useful signals about safety and quality, they cannot fully capture real-world performance. As these systems scale, ongoing monitoring for adverse events and near-miss unsafe recommendations will remain important.

Future work can extend this foundation in several directions. Clinically, longer and broader pragmatic trials across additional conditions will be necessary. These should include, for example, hypertension, hyperuricemia and musculoskeletal rehabilitation. Trials should also span diverse delivery settings, such as community programs, outpatient clinics, and employer wellness programs. Together, these studies will help estimate overall effectiveness, heterogeneity of treatment effects, and cost-effectiveness. Methodologically, future work could strengthen safety by separating generation from verification: an LLM-based safety checker can screen draft feedback for contraindications, excessive intensity, and missing escalation triggers, then either revise to a safer alternative or route the case to human review. From a deployment perspective, designing tiered oversight—where high-risk cases trigger stricter review and low-risk cases use lightweight verification—may enable safe scaling without overwhelming human reviewers. Finally, participant-centered design should remain central: our participant-reported results suggest that tone, encouragement, and perceived emotional stability are potential drivers of adherence and outcomes, and thus deserve the same rigor in measurement and optimization as biomedical endpoints.

In summary, an expert-aligned LLM can improve the scalability and effectiveness of digital exercise coaching when deployed as part of an EPS–human collaboration workflow. By combining domain-specific alignment, prospective evaluation, and accountable human oversight, this study provides a practical template for translating LLM capabilities into measurable benefit for real-world chronic disease management.

4 Methods

4.1 Ethics approval

The study protocol covered both the pilot study and the randomized controlled trial, and was reviewed and approved by the Ethics Review Committee of City University of Hong Kong and other participating centers. The trial was prospectively registered at the Chinese Clinical Trial Registry (identifier ChiCTR2600118939). Written informed consent was obtained from all participants before enrollment. All procedures were

conducted in accordance with the Declaration of Helsinki and, when applicable, the International Council for Harmonisation Good Clinical Practice (ICH-GCP) guidelines. Participants were not informed of their individual arm assignment. The system was used within a prespecified oversight and safety framework and was not intended to replace clinical judgment or serve as the sole basis for medical decision-making. Study data were de-identified and stored using access-controlled, encrypted systems, and only authorized study personnel had access to identifiable information.

4.2 Overall study design

As shown in Fig. A1, this study comprised four tightly coupled components: data curation, model development, internal validation and clinical trial. The first section describes data curation, including the collection and processing of question-answer pairs from multiple sources and the construction of expert-curated exercise guidance entries. The second section describes the model development pipeline, encompassing supervised fine-tuning and preference alignment using Kahneman-Tversky Optimization (KTO) to build the Exercise Prescription System (EPS). The third section describes the internal validation of EPS performance through established medical benchmarks and a pilot study with multidimensional evaluation criteria. The fourth section describes the randomized controlled clinical trial comparing the EPS–human arm with the Human arm in a 21-day health intervention program.

4.3 Dataset construction

4.3.1 Public exercise and weight-management corpus (D1)

To construct the public exercise and weight-management corpus (D1), we assembled a pool of publicly available medical instruction datasets and performed bilingual keyword-based selection to isolate exercise- and weight-management–relevant instances [27]. We curated a controlled vocabulary containing both Simplified Chinese strings and their English counterparts because candidate sources included both Chinese- and English-language questions and answers. We retained an example if any keyword appeared in either the question or the reference answer after light text normalization. Matching was case-insensitive for English and used substring matching for Chinese. The full keyword lexicon and matching rules are provided in Supplementary Table 1. We then removed exact duplicates and standardized formatting across sources. Table B1 summarizes the composition of D1 by target application, language and number of examples.

4.3.2 Expert-curated exercise prescription dataset (D2)

We constructed the expert-curated exercise prescription dataset (D2) to provide high-quality alignment signals for EPS in exercise counseling tasks. D2 comprised four components, including population-level exercise guidance, user-personalized exercise suggestions, expert exercise reviews and an exercise type database (Table C2). To build D2, we used GPT-5 to generate draft content for all components. Professional health managers then reviewed each instance item by item, correcting unsafe

or non-actionable content, aligning recommendations with routine coaching practice and ensuring that the final outputs were feasible and clinically appropriate. Exercise-feedback drafts were generated using a standardized SOAP-based prompt specification that was co-developed by five domain experts, and the SOAP prompt template is provided in Supplementary Table 2. This human-in-the-loop curation ensured that all outputs were reviewed and approved by experts.

4.4 Model architectures and training procedure

4.4.1 Base language models

We selected two open-weight model families as the base language models for EPS: Qwen3 (8B and 14B parameters) and DeepSeek-R1 (8B and 14B parameters). This choice balanced strong general instruction-following and bilingual capability with practical efficiency for iterative domain adaptation and deployment in real-world workflows.

Qwen3 is a multilingual decoder-only transformer model series that spans multiple dense model sizes and integrates reasoning-oriented thinking behavior with standard response generation in a unified framework [28]. DeepSeek-R1 is a reasoning-focused model trained via a multi-stage framework that combines reinforcement learning and alignment stages, with smaller dense variants distilled to retain reasoning capability at lower computational cost [29]. All base models were obtained from their respective official repositories and used under their respective licenses. Qwen3 models were used under the Apache-2.0 license, whereas DeepSeek-R1 and its derived variants were used under the MIT license.

4.4.2 Supervised fine-tuning

As the first stage of model development, we performed supervised fine-tuning (SFT) on the public exercise and weight-management corpus (D1) to adapt the base models to medical and exercise-counseling instructions and responses. SFT was performed with parameter-efficient LoRA [30] adaptation, implemented using the LLaMA-Factory training framework. We trained with a maximum sequence length of 3,072 tokens for 14B models and 4,096 tokens for 8B models, using a cosine learning-rate schedule with 20 warmup steps. The learning rate was set to 5×10^{-5} and models were trained for 5 epochs. The per-device batch size was 1 (14B) or 2 (8B) with gradient accumulation (16 for 14B and 8 for 8B). We held out 10% of the training data for validation and performed evaluation every 500 steps (14B) or every 50 steps (8B), saving checkpoints periodically and selecting the best checkpoint based on validation performance. Training used mixed precision (bf16 for 14B and fp16 for 8B) and enabled gradient checkpointing where supported.

4.4.3 Preference alignment with Kahneman–Tversky Optimization

After supervised fine-tuning on D1, we further aligned the models with preference signals using Kahneman–Tversky Optimization (KTO), a prospect-theory–motivated objective that learns from a binary desirability signal rather than pairwise rankings

[31]. We constructed KTO training data from the four components of D2. For each prompt, GPT-5 generated candidate outputs, which professional health managers reviewed item by item. Reviewers assigned each output a desirable or undesirable label and revised unsafe or non-actionable content when needed. This workflow coupled LLM generation with expert oversight and iterative refinement.

KTO training was implemented in LLaMA-Factory with LoRA adapters. We used a maximum sequence length of 4,096 tokens, cosine learning-rate scheduling, a 10% validation split, and selected the best checkpoint by validation performance. For 8B models we trained with a per-device batch size of 2, gradient accumulation of 16, and bf16 precision, whereas for 14B models we used gradient accumulation of 8 with fp16. Learning rates were set to 5×10^{-7} (DeepSeek-R1–8B), 2×10^{-6} (Qwen3–8B), and 1×10^{-6} (both 14B variants), and we evaluated and saved checkpoints every 25 steps.

Unless otherwise specified, the EPS used in both the pilot study and the randomized controlled trial was a fine-tuned Qwen3-14B model (EPS-Qwen3-14B). The model checkpoint and inference configuration were fixed before study initiation.

4.5 Benchmark construction and evaluation protocol

We selected four established medical question-answering benchmarks for internal validation, including two Chinese datasets (CMB and CMExam) and two English datasets (MedMCQA and MedQA), and Table D reports the resulting test-subset sizes. We did not use the full original benchmarks. Instead, we constructed filtered test subsets by applying the bilingual keyword lexicon (Supplementary Table 1) to each benchmark test set and retaining an item if any keyword appeared in the question text or the reference answer [27]. We removed exact duplicates and verified that the filtered benchmark items did not overlap with the training corpora used for fine-tuning.

Benchmark evaluation followed an automatic, single-best-answer multiple-choice protocol. For each item, we prompted the model to analyze the options and output a single option letter on the final line using a fixed answer tag, which enabled deterministic parsing and exact-match scoring. We evaluated the fully fine-tuned checkpoints by loading the corresponding base model and sequentially merging the SFT and KTO LoRA adapters before inference. Decoding used top-p sampling with a fixed configuration. We set the temperature to 0.35, top-k to 40, top-p to 0.8, the repetition penalty to 1.15, and the maximum number of new tokens to 512. To account for sampling variability, we repeated inference 10 times with different random seeds and reported the mean accuracy across runs. This benchmark suite primarily assesses constrained medical knowledge and safety-relevant reasoning, whereas richer judgments of real-world exercise-feedback quality were assessed in the subsequent expert pilot study, consistent with prior prospective evaluations that complemented automatic metrics with human ratings.

4.6 Expert evaluation study design

We conducted an internal expert evaluation to assess the quality of exercise feedback and to quantify the incremental effects of domain supervision and preference alignment before clinical deployment. The evaluators had a median of 4.7 years of professional

experience in exercise and lifestyle coaching (IQR, 2.9–6.8), and were not involved in model training or prompt engineering. Using the same Qwen3-14B backbone, we compared three models: the base model, the model after supervised fine-tuning on D1, and EPS after preference alignment. Exercise feedback was generated for a set of de-identified user profiles and corresponding exercise records that reflected routine coaching scenarios. For each case, the model was instructed to generate feedback following a standardized SOAP-based format.

Professional health managers independently rated the responses using a structured questionnaire administered via WeChat with Tencent Questionnaire. Responses from different models were presented in randomized order without system identifiers. We summarized the rating distributions and aggregated scores for each dimension to compare the three models. Each dimension was rated using a four-level rubric (A–D) mapped to a 0–3 score. We summarize scores as means with 95% confidence intervals (CIs) and report two-sided P values for pairwise comparisons.

4.7 Real-world deployment and clinical trial design

4.7.1 Study setting and participants

The randomized controlled trials were embedded in WeChat-based online chronic disease management communities operated by Ping An Health and Technology Co., Ltd. All lifestyle coaching was delivered by a centralized team of professional health managers employed by the company and based in Shenzhen, China. Health managers had a median of 3.2 years of experience in weight-management and chronic-disease lifestyle coaching (IQR, 1.8–5.1; range, 0.5–9.6 years). They completed standardized internal training on exercise prescription, risk escalation and WeChat-based community delivery, and the same staffing pool served both trial arms. Participants originated from the company's life-insurance user base. Most participants resided in mainland China and a small proportion resided in Hong Kong. The deployment and trial procedures followed the WHO guidance on ethical considerations for AI in health, with an emphasis on preserving clinical validity and maintaining human oversight in care delivery. All study workflows and data handling were designed to meet full Health Insurance Portability and Accountability Act (HIPAA) compliance. Health managers assessed eligibility, obtained informed consent, and enrolled participants before 1:1 random assignment.

Participants were assigned to either the Human arm or the EPS–human arm. Participants in the EPS–human arm received exercise feedback drafted by EPS-Qwen3-14B and reviewed by health managers before delivery, whereas participants in the Human arm received feedback written entirely by health managers. The trials included a weight-loss cohort and a glycemic-control cohort, and participant flow and baseline characteristics are reported in the main text.

4.7.2 Intervention and control conditions

The randomized controlled trial used a single-blind design. Participants were not informed of their arm assignment, and exercise feedback was delivered through the same WeChat community workflow in both arms; the only difference was whether the draft feedback was generated by EPS and reviewed by a health manager or written

entirely by the health manager. Health managers were necessarily unblinded because they performed message delivery in both arms and reviewed EPS drafts in the EPS–human arm. For analysis, arm labels were replaced with anonymized group codes, and the statistical analysts remained blinded until the primary analyses were finalized.

Both cohorts followed a standardized 21-day WeChat-based weight-management program delivered through online communities. The program workflow was prespecified in an operational SOP and was identical across study arms apart from how exercise feedback was drafted. Core components included structured daily check-ins and reminders, group-based education messages, and health-manager follow-up. Participants were prompted to report body weight and to share meal photos and logs for routine dietary coaching, and health managers provided corresponding comments and encouragement. The program also included scheduled onboarding and engagement activities, weekly wrap-up messages, and a graduation ceremony. Weekly educational themes progressed from a balanced-diet module in week 1 to exercise promotion in week 2 and habit-formation and emotion-focused strategies in week 3. After participants posted exercise records in the WeChat community, exercise feedback was delivered as a reply in the same thread and the participant was @-mentioned to trigger an in-app notification, using the same delivery workflow in both arms.

Participants assigned to the EPS–human arm received exercise feedback drafted by EPS-Qwen3-14B from the participant profile and the corresponding activity record. Professional health managers reviewed each draft before delivery and could edit it to correct unsafe or non-actionable content, with the final message delivered by health managers under human accountability. Participants assigned to the Human arm received exercise feedback written entirely by health managers within the same WeChat workflow and service scope.

4.7.3 Outcome measures

We prespecified outcome measures to capture short-term clinical effectiveness, operational mechanisms of delivery, and participant-reported experience during the standardized 21-day program delivered in WeChat-based chronic disease management communities.

For the weight-loss cohort, the prespecified primary outcome was absolute change in body weight (kg) from baseline (day 0) to the end of the 21-day program (day 21). Prespecified secondary outcomes included percent change in body weight and the proportion of participants achieving clinically meaningful weight loss (defined as $\geq$2% reduction from baseline). For the glycemic-control cohort, the prespecified primary outcome was absolute change in fasting glucose (mmol/L) from baseline to day 21, and a prespecified secondary outcome was the fasting-glucose reduction ratio (percent change from baseline).

To quantify the hypothesized delivery mechanism of EPS–human collaboration, we additionally measured the number of exercise-feedback instances delivered over the 21-day intervention window, extracted from the routine coaching workflow logs.

Participant-reported outcomes were collected after program completion using a prespecified questionnaire assessing satisfaction and perceived affective and motivational qualities of the exercise feedback.

Outcome selection and grouping were aligned with prior pragmatic evaluations of model-assisted clinical workflows that combined objective operational or clinical endpoints with structured questionnaires capturing patient and clinician experience [17, 18].

4.7.4 User questionnaire design and data processing

At program completion, participants in both arms were invited to complete an end-of-program questionnaire assessing their experience with the exercise-feedback component. Participants received a RMB 10(approximately USD 1.4) incentive upon completion of the end-of-program questionnaire to encourage participation. The incentive was offered identically in both trial arms and was not contingent on questionnaire responses. To support participant masking, the end-of-program questionnaire used neutral item stems that did not reference the feedback source, such as EPS or health managers. The questionnaire was administered in Chinese and consisted of 15 items (Supplementary Table 4). Item 1 served as a screening question asking whether the participant had used or seen the exercise feedback. Items 2–15 comprised 14 statements rated on a 7-point Likert scale (1 = strongly disagree to 7 = strongly agree).

Items 2–15 were organized into prespecified experience domains. Items 2–4 assessed overall satisfaction and interaction experience. Items 5–10 assessed perceived emotional stability and tone consistency, covering calmness and stability as well as cross-context consistency. Items 11–15 assessed perceived agreement and affirmation style, capturing perceived agreement and praise or ingratiation.

We excluded responses from participants who reported never having seen the exercise feedback (screening item 1), responses with implausibly short or long completion times, and straight-lined responses in which the same option was selected for all items. In the Human arm, we received 453 questionnaires, of which 412 respondents indicated that they had seen the exercise feedback in item 1; after applying the remaining quality-control filters, 344 questionnaires were retained for analysis. In the EPS–human arm, we received 478 questionnaires, of which 427 respondents indicated that they had seen the exercise feedback in item 1; after applying the remaining quality-control filters, 336 questionnaires were retained for analysis.

We analyzed item-level scores (Items 2–15) and summarized each item as mean with two-sided 95% confidence intervals. Between-arm comparisons were performed using two-sided Welch's t-tests (unequal variances), consistent with common practice in pragmatic evaluations that compare questionnaire-based experience measures across trial arms.

4.8 Statistical analysis

Analyses were conducted with masked group labels, such as Group A versus Group B. Unmasking was performed only after the primary analysis scripts and results were finalized.

All analyses were conducted in Python. Continuous variables were summarized as mean (s.d.), and figures report means with 95% confidence intervals. Binary outcomes

were summarized as proportions with 95% confidence intervals. All tests were two-sided with $\alpha = 0.05$. Analyses used participants included in the final analysis sets shown in the CONSORT flow diagrams. No outcome imputation was performed, and each endpoint was analyzed using available data.

Baseline continuous variables were compared using two-sided Welch's unequal-variance t-tests. Baseline categorical variables were compared using Pearson's χ^2 test or Fisher's exact test when expected counts were small. In the weight-loss cohort, between-arm contrasts for weight change and percent weight change used Welch's t-tests. We reported unadjusted mean differences (EPS–human minus Human) with Welch-based 95% confidence intervals. The $\geq$2% weight-loss rate was compared using a two-proportion z-test. We reported arm-level event rates with exact Clopper–Pearson 95% confidence intervals, and summarized between-arm effects as risk differences and relative risks. A prespecified sensitivity analysis estimated adjusted mean differences using ordinary least squares regression with baseline covariates. In the glycemic-control cohort, within-arm changes were summarized as mean changes with 95% confidence intervals, and between-arm contrasts used Welch's t-tests with 95% confidence intervals. Because the glycemic-control cohort was small ($n = 40$ in the final analysis), we performed a nonparametric bootstrap sensitivity analysis to assess robustness to distributional assumptions. We estimated 95% bootstrap confidence intervals for within-arm mean changes and between-arm mean differences (EPS–human minus Human) using $B = 10{,}000$ resamples (percentile intervals) with a fixed random seed for reproducibility. Primary hypothesis tests and P values remained those of the prespecified two-sided Welch's t-tests, and the bootstrap analysis was used to quantify uncertainty and confirm robustness. Expert pilot outcomes and end-of-program questionnaires used nonparametric and Welch-based tests, respectively, with Holm step-down adjustment for multiple testing. Benchmark performance was summarized as accuracy on the filtered subsets and averaged across repeated runs with different random seeds.

5 Data availability

6 Code availability

7 Acknowledgements

8 Author contributions

9 Competing interests

10 Additional information

Appendix A Study design overview

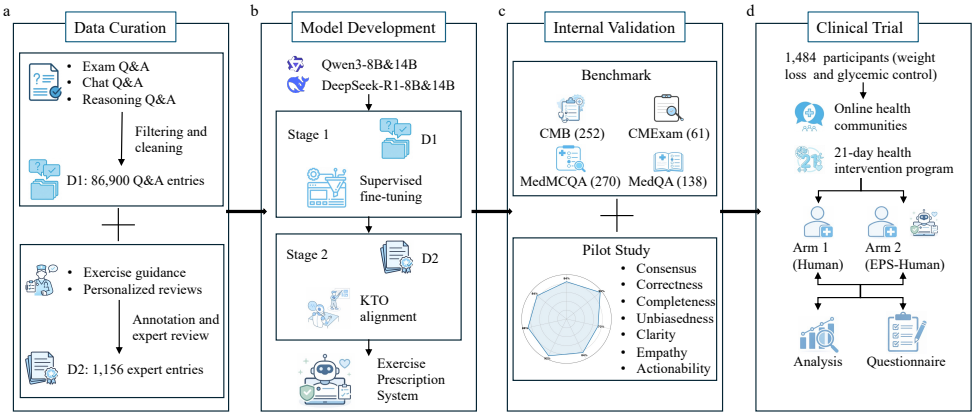

Fig. A1 Study design overview. **a**, Data curation process. Question-answer pairs were collected from examination databases, chat logs, and reasoning tasks, then filtered and cleaned to construct the D1 dataset (86,900 entries). Additionally, exercise guidance and personalized review data underwent expert annotation to form the D2 dataset (1,156 entries). **b**, Model development pipeline. Base models (Qwen3-8B&14B and DeepSeek-R1-8B&14B) underwent two-stage training: supervised fine-tuning with D1 followed by KTO alignment with D2, yielding the Exercise Prescription System (EPS). **c**, Internal validation. Model performance was evaluated on four filtered medical benchmarks (CMB, CMExam, MedMCQA, and MedQA), and a pilot study assessed response quality across seven dimensions: consensus, correctness, completeness, unbiasedness, clarity, empathy and actionability. **d**, Clinical trial design. A total of 1,484 participants targeting weight loss and glycemic control participated in online health communities and were randomly assigned to either the human arm or the EPS-human arm for a 21-day health intervention program, followed by outcome analysis and questionnaire assessment.

Appendix B Composition of the public training corpus (D1)

Table B1 Composition of the public training corpus (D1).

Target application	Dataset	# Examples	Language
Multiple-choice QA	MedBooks-18-CoT [32]	11654	English
Miscellaneous	MedInstruct-52K [33]	7999	English
Free-form/Single-turn QA	ChatDoctor [34]	11738	English
Free-form/Single-turn QA	MedicationQA [35]	17	English
Clinical Note Generation	MTS-Dialog [36]	71	English
Free-form/Single-turn QA	LiveQA [37]	69	English
Multiple-choice QA	MedMCQA_train [38]	1510	English
Multiple-choice QA	MedQA_train [39]	1079	English
Multiple-choice QA	CMB_train [40]	1225	Chinese
Multiple-choice QA	MultiMedQA [20]	49	English
Multiple-choice QA	CMExam_train [41]	265	Chinese
Multiple-choice QA	Medbullets [42]	98	English
Free-form/Single-turn QA	Huatuo_EncyclopediaQA [43]	37271	Chinese
Knowledge-graph QA	Huatuo_KnowledgeGraphQA [43]	6349	Chinese
Free-form/Single-turn QA	Huatuo26M-Lite [43]	6948	Chinese
Complex-reasoning medical QA	medical-o1-reasoning-SFT [44]	558	Chinese

Appendix C Composition of the private training corpus (D2)

Table C2 Composition of the private training corpus (D2).

Dataset	**# Examples**
Population-level exercise guidance	56
User-personalized exercise suggestions	164
Expert exercise reviews	856
Exercise type database	80

Appendix D Benchmark

Benchmark	# Examples	Language
CMB_test [40]	252	Chinese
CMExam_test [41]	61	Chinese
MedMCQA_test [38]	270	English
MedQA_test [39]	138	English

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